



03

Image Processing 1

Digital image processing

Digital image processing is the use of a digital computer to process digital images through an algorithm. As a subcategory or field of digital signal processing, digital image processing has many advantages over analog image processing. It allows a much wider range of algorithms to be applied to the input data and can avoid problems such as the build-up of noise and distortion during processing.

In particular, digital image processing is a concrete application of, and a practical technology based on: Classification, feature extraction, multi-scale signal analysis, pattern recognition, projection

Some techniques which are used in digital image processing include: Hidden Markov models, Image editing, Image restoration, Principal component analysis, Wavelets, Etc..

Some Applications of Digital Image Processing

- Digital camera images
- Film
- Face detection
- Improvement of image quality method



Some Tasks of Image Processing

- Color mapping
- Blending & parsing
- Image threshold
- Blurring & smoothing
- Morphological operator
- Gradients
- Histograms

Color mapping

Color mapping (photography) is a function that maps (transforms) the colors of one (source) image to the colors of another (target) image. A color mapping may be referred to as the algorithm that results in the mapping function or the algorithm that transforms the image colors. Color mapping is also sometimes called color transfer or, when grayscale images are involved, brightness transfer function (BTF); it may also be called photometric camera calibration or radiometric camera calibration.

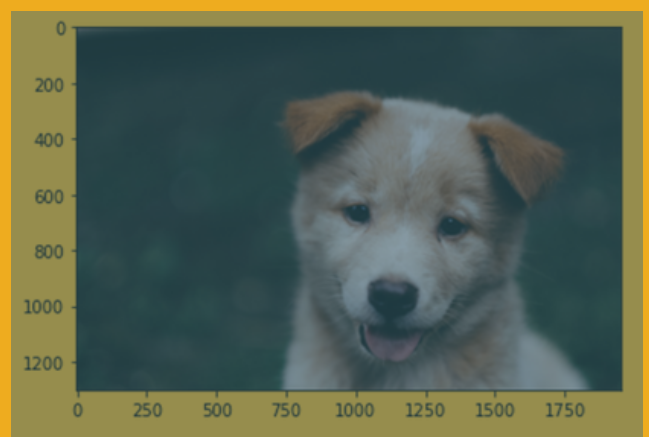
When the image file is read with the OpenCV function `imread()`, the order of colors is BGR (blue, green, red). On the other hand, in Pillow, the order of colors is assumed to be RGB (red, green, blue). Therefore, if you want to use both the Pillow function and the OpenCV function, you need to convert BGR and RGB.

```
img = cv2.imread('../DATA/00-puppy.jpg')  
plt.imshow(img)
```



Before

```
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)  
plt.imshow(img)
```



After

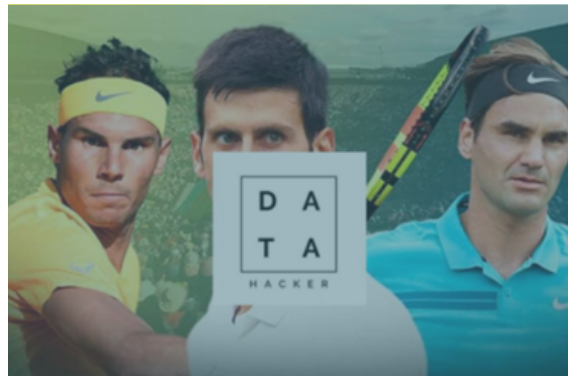
BGR -> RGB

Blending and Parsing

Blending and pasting is used commonly to mix two images or overlay an image on top of another image. There are many applications for blending and pasting such as graphic designing, logo placement, internet meme generation. All of these applications can be easily and quickly processed using OpenCV.

The image pasting is a process of overlaying one image on top of the other. To successfully apply this process in OpenCV we need to select Region of Interest (ROI) in the first image, and then apply masking and some logical operations to overlay second image over the first image.

Pasting image



The process of blending done through the `cv2.addWeighted()` function from OpenCV.

This function uses both the input images, assigns certain weights to each image pixel, adds them together and outputs the result into a new pixel. In that way we will get the us an impression of transparency. In the following example you can see the difference between pasting one image on top of other and blending two images together.

Blending image

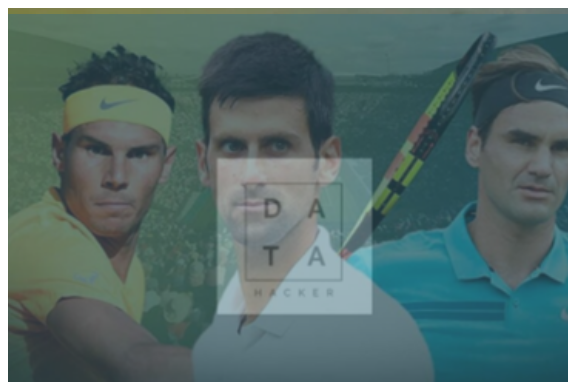


Image Threshold

Thresholding is a type of image segmentation, where we change the pixels of an image to make the image easier to analyze. Most frequently, we use thresholding as a way to select areas of interest of an image, while ignoring the parts we are not concerned with.

There are several types of threshold for image processing:

- Binary
- Binary inverse
- Threshold truncation
- Threshold to zero
- Inverse threshold to zero

Example of threshold



Picture Source

- unsplash.com
- opencv.org
- shutha.org
- mobidev.biz
- pexels.com